Q2 2025 Earnings Call

Company Participants

- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, Chairman, President and Chief Executive Officer
- Satya Kumar, Corporate Vice President, Investor Relations and Treasury

Other Participants

- C. J. Muse, Cantor Fitzgerald
- Chris Caso, Wolfe Research
- Christopher Danely, Citigroup
- Harlan Sur, J. P. Morgan Chase & Co.
- Joseph Moore, Morgan Stanley
- Krish Sankar, TD Cowen
- Timothy Arcuri, UBS

Presentation

Operator

Thank you for standing by and welcome to Micron's Second Quarter 2025 Financial Call. At this time, all participants are in a listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions). As a reminder, today's program is being recorded.

And now, I'd like to introduce your host for today's program, Satya Kumar, Corporate Vice President, Investor Relations and Treasury. Please go ahead, sir.

Satya Kumar {BIO 18241786 <GO>}

Thank you, and welcome to Micron Technology's fiscal second quarter 2025 financial conference call.

On the call with me today are Sanjay Mehrotra, our Chairman, President and CEO; and Mark Murphy, our CFO.

Today's call is being webcast from our Investor Relations site at investors.micron.com, including audio and slides. In addition, the press release detailing our quarterly results has been posted on the website along with prepared remarks for this call.

Today's discussion of financial results is presented on a non-GAAP financial basis unless otherwise specified. A reconciliation of GAAP to non-GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on financial conferences that we may be attending. You can also follow us on X at MicronTech.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, including demand for our products, our market share, market pricing, and cost trends and drivers, our plans for manufacturing, the impact of developing technologies such as AI, product ramp plans, technologies and market position, expected capabilities of our future products, our planned investments and expenditures, our expected results and guidance, regulatory matters, and other matters.

These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from statements made today. We refer you to documents we filed with the SEC, including our Form 10-K, Forms 10-Q, and other reports and filings for a discussion of risks that may affect our future results. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance, or achievements. We are under no duty to update any of the forward-looking statements to confirm these statements to actual results.

I will now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Satya. Good afternoon, everyone.

Micron is in the best competitive position in our history, and we are achieving shared gains across high-margin product categories in our industry. Our strong product momentum has enabled us to build deeper customer relationships, and Micron's industry-leading products are now more firmly entrenched in our customers' high-value product roadmaps.

In fiscal Q2, data center DRAM revenue reached a new record. HBM revenue grew more than 50% sequentially to a new milestone of over $1 billion of quarterly revenue. Our HBM shipments were ahead of our plans, demonstrating strong execution of our ongoing ramp. The combination of our revenue from high-capacity DRAM modules and our industry-leading LP DRAM for the data center also exceeded the $1 billion milestone for the quarter.

Micron remains the only company in the world to ship low-power DRAM into the data center in high volume, showcasing our pioneering innovation and deep partnership with our customers for differentiated solutions. As we build on this momentum, we expect fiscal Q3 revenue to be another record for Micron, driven by shipment growth across both DRAM and NAND. We see the combination of AI data center demand and the ramp of HBM and its associated trade ratio contributing to tightness at the leading edge and constraining non-HBM DRAM supply. We expect supply actions announced by NAND companies to improve the dynamics in the NAND market.

Micron's 1-beta DRAM technology leads the industry, and we are extending our leadership with the launch of our 1-gamma node, and the industry's first shipments of 1-gamma-based D5 products last month. Micron's 1-gamma is our first DRAM node incorporating EUV, and we have achieved 20% lower power, 15% better performance, and over 30% improvement in bit-density compared to our 1-beta DRAM. Micron's leading edge Gen 9 NAND technology node delivers the industry's fastest TLC-based NAND, and we are managing the ramp of this node mindful of the supply-demand balance in the industry.

Micron continues to make disciplined investments that position us to capitalize on the significant growth opportunities driven by AI. We are focused on growing HBM capacity in our existing manufacturing facilities to meet requirements through 2026. In January, we broke ground on an HBM advanced packaging facility in Singapore. This investment allows us to meaningfully expand our total advanced packaging capacity beginning in calendar 2027.

Our new DRAM fab construction in Idaho completed an important construction milestone that enabled the receipt of the first disbursement of funding from our CHIPS grant for the project during the quarter. This new Idaho fab will provide meaningful DRAM output starting in fiscal 2027.

Turning to our end markets. Dramatic improvements in computation hardware have reduced the per-token cost of generative AI models. These hardware improvements, along with more efficient algorithms and software, drive down the cost of inference and make generative-AI-based capabilities more accessible to new applications and use cases. This broadening deployment creates a powerful growth vector for aggregate AI demand and recent innovations and those in the pipeline from key contributors to the AI ecosystem will continue to fuel this growth trend.

As GPU and custom AI accelerator performance capabilities continue to improve with each new generation of product, these high-performance processors are starved of memory bandwidth. HBM memory provides the bandwidth necessary to leverage these powerful processors in the most effective and efficient manner, and we are excited to see the growth opportunities ahead for this complex and high-value product category, where our customers now recognize Micron as the HBM technology leader in our industry.

Recently, large hyperscale customers reiterated strong year-over-year growth for their capital investments in calendar 2025. We project mid-single-digit server unit growth in calendar 2025, with growth in both traditional and AI servers.

We see strong demand for HBM and have once again increased our HBM TAM estimate for calendar 2025 to over $35 billion. We remain on track to reach HBM shares similar to our overall DRAM supply share on a run rate basis in calendar Q4 2025. As previously mentioned, Micron is sold out of our HBM output in calendar 2025. We are seeing strong demand for our HBM supply in 2026 and are in discussions with our customers on agreements for their calendar 2026 HBM demand.

Micron's industry-leading HBM3E delivers a 30% power reduction compared to the competition, and our HBM3E 12-high has a remarkable 20% power advantage over competing 8-high products, while providing a 50% higher memory capacity. We have begun volume production of HBM3E 12-high and are focused on ramping capacity and yield. We anticipate HBM3E 12-high will comprise the vast majority of our HBM shipments in the second half of calendar 2025.

We are making good progress on additional platform and customer qualifications with HBM. Micron's HBM3E 8-high is designed into NVIDIA's GB200 system and our HBM3E 12-high is designed into the GB300. In fiscal Q2, we initiated volume shipments to our third large HBM3E customer and anticipate additional customers over time. We expect multi-billion dollars in HBM revenue in fiscal 2025.

Looking ahead, we are enthusiastic about Micron's HBM4, which will ramp in volume in calendar 2026. Our HBM4 provides a bandwidth increase of over 60% compared to HBM3E. The timing of our HBM4 is aligned to our customers' requirements, and we are focused on delivering the best HBM4 products to the market across power efficiency, quality, and performance. Our proven HBM product performance, our strong HBM roadmap, and our demonstrated manufacturing excellence uniquely position Micron to capitalize on next-generation HBM4 and HBM4E solutions.

Micron has led the adoption of LP in data centers. In AI servers, Micron's LP lowers memory power consumption by over two-thirds compared to D5. We expect to maintain our leadership position in LP for server as it transitions from soldered components to a SOCAMM, or Small Outline Compression Attached Memory Module form factor. Micron's SOCAMM was developed in collaboration with NVIDIA to support the GB300. LP DRAM in a SOCAMM form factor enables easier server manufacturability and serviceability and helps drive broader LP adoption in the server market. We are on track to deliver multi-billion dollars in revenue in fiscal 2025 from our portfolio of high-capacity D5 modules and LP products for the data center.

In data center NAND, demand moderated in fiscal Q2 due to short-term customer inventory-driven impacts, and we see a return to bit shipment growth in the months ahead. In calendar Q4 2024, based on industry analyst reports, Micron achieved yet another record high market share in data center SSDs, with revenue growth in each category, including performance, mainstream, and capacity SSDs. Our high-performance 9550 SSD, which is on NVIDIA's GB200 NVL72 approved vendor list, completed qualifications at multiple customers'.

During the quarter, we announced that Micron's data center-class G8 QLC-based NAND components are qualified for production in Pure Storage's high-capacity 150-terabyte direct flash module. Micron's data center-class NAND components give customers the ability to leverage our industry-leading NAND design and process technology in their custom storage solutions. Micron's leadership in QLC NAND supports the transition from HDD to NAND solutions in the data center. We expect to generate multiple billions of dollars in data center NAND revenue, and once again, grow our data center NAND market share in calendar 2025.

We expect the PC market to grow mid-single-digits in unit terms in calendar 2025, with growth weighted to the second half of calendar 2025. The Windows 10 end of life in October 2025, combined with an aging installed base and the desire amongst customers to ensure that their PC hardware specs can support compelling AI applications in the future, are key catalysts that drive this growth.

AI PCs require a minimum of 16 gigabytes of DRAM, with many models requiring even higher memory, versus the average 12-gigabyte PC content last year. During the quarter, we sampled our 16-gigabit 1-gamma-based D5 products to PC clients. In NAND, we launched our Gen 9-based 4,600 performance SSDs, the fastest in the world for the client market, and completed qualifications of our 2,650 mainstream SSDs at multiple PC OEMs.

Turning to mobile, our expectations for smartphone unit volume growth in calendar 2025 remain at low-single-digit percentages. Smartphone customer inventory dynamics have played out as anticipated, leading to mobile DRAM and NAND bit shipment growth in our fiscal Q3. AI adoption continues to be a significant driver for increased mobile DRAM demand. AI-capable flagship phones increasingly feature DRAM capacities of 12-gigabyte or higher, compared to the 8-gigabyte in last year's models.

Smartphone OEMs are using Micron's industry-leading 9.6-gigabit-per-second LP5X DRAM to improve AI performance, delivering up to 20% more tokens per second than those using legacy speed grades on the same SoC. During the quarter, we announced that our LP5X DRAM and UFS 4.0 NAND were featured in the high-end of the Samsung Galaxy S25 series. Micron's mobile DRAM and UFS storage solutions are in high demand and will continue to launch in flagship and high-end smartphones throughout the year. Additionally, we are now sampling the industry's first mobile G9 managed NAND-based UFS 4.1 solution in densities up to 1 terabyte.

Automotive OEMs, industrial, and consumer-embedded customers are in the later stages of adjusting their inventory levels. In automotive, which comprises the largest portion of our EBU revenue, memory and storage content per car continues to increase as AI-enabled in-vehicle infotainment systems become more enriched and driver assistance functions become more capable. Advanced robo-taxi platforms today contain over 200 gigabytes of DRAM, or 20x to 30x higher than the amount of DRAM in the average car.

Micron is well-positioned to capitalize on this trend with our industry-leading portfolio of automotive products. During the quarter, we announced the production readiness of the industry's first automotive LP5X DRAM product that supports a 9.6 gigabit per second speed grade, addressing the increasing performance requirements of AI-driven applications in vehicles. Additionally, our 4150 SSD became the industry's first enterprise SSD product that is automotive-qualified and is now sampling at target customers, further reinforcing our commitment to innovation and leadership in this important market.

Now, turning to our market outlook. Calendar 2024 DRAM bit demand growth was in the high-teens, consistent with our prior expectations. Calendar 2024 NAND bit demand growth was approximately 10%, slightly below our previous view of low-double-digits. We forecast calendar 2025 DRAM bit demand growth in the mid to high-teens percentage range and NAND in the low-double-digit percentage range. Over the medium-term, we expect industry bit demand growth of mid-teens CAGR for both DRAM and NAND.

As we have previously discussed, NAND technology transitions provide a significant increase in overall bit output. Sustained NAND industry supply-demand balance can result from increasing the time between node transitions along with sustained reductions in NAND industry CapEx and wafer capacity. NAND industry wafer capacity underutilization can help to improve the near-term dynamics in the NAND market.

We expect Micron's supply growth in calendar 2025 to be lower than industry demand growth for both DRAM and NAND. We expect our inventory days to decline as we move through calendar 2025. We expect to maintain our bit share in DRAM and NAND in calendar 2025.

In DRAM, we expect a strong ramp of HBM throughout calendar 2025. As noted before, HBM3E consumes 3x the amount of silicon compared to D5 to produce the same number of bits. Looking ahead, we expect the trade ratio to increase with HBM4 and then again with HBM4E when we expect it to exceed 4:1. This sustained and significant increase in silicon intensity for the foreseeable future contributes to tightness for industry-leading-edge node supply and constrains capacity for non-HBM products.

In NAND, we continue to underutilize our fabs and our wafer output is down mid-teens percentage from prior levels. We plan to reuse a portion of our underutilized NAND equipment to support capital-efficient conversions to leading-edge nodes. This strategy results in over 10% structural reduction of NAND wafer capacity exiting fiscal 2025 compared to levels exiting fiscal 2024. We will continue to prudently manage our NAND supply, including the levels of our capital investment, the pace of ramp of our new technology node, fab capacity, and utilization consistent with our demand growth.

Our capital spending plans remain unchanged at approximately $14 billion for fiscal 2025. A significant portion of our capital investments are focused on multi-year facility investments to support our DRAM and HBM manufacturing, including our Idaho fab, Singapore HBM advanced packaging facility, and Taiwan DRAM test facility. Micron will remain disciplined with our overall equipment investments to manage our supply growth consistent with demand.

On tariffs, Micron serves as the U.S. importer of record for a very limited volume of products that would be subject to newly announced tariffs on Canada, Mexico, and China. We continue to monitor the possibility of future tariffs and are prepared to work with our customers and suppliers to understand future tariff effects and supply chain options that may arise. Where tariffs do have an impact, we intend to pass those costs along to our customers.

With that, I will now turn it over to Mark for our financial results and outlook.

Mark Murphy {BIO 16136048 <GO>}

Thank you, Sanjay, and good afternoon, everyone.

Micron delivered fiscal Q2 EPS above the guidance range and revenue and gross margin within the range. Total fiscal Q2 revenue was approximately $8.1 billion, down 8% sequentially and up 38% year-over-year.

Fiscal Q2 DRAM revenue was $6.1 billion, up 47% year-over-year, and represented 76% of total revenue. Sequentially, DRAM revenue decreased 4%, with bit shipments decreasing in the high-single-digit percentage range and prices increasing in the mid-single-digit percentage range as a result of improving portfolio mix.

Fiscal Q2 NAND revenue was $1.9 billion, up 18% year-over-year, and represented 23% of Micron's total revenue. Sequentially, NAND revenue decreased 17%, with bit shipments modestly higher and prices decreasing in the high-teens percentage range. Fiscal Q2 NAND bit shipments were above our expectations, driven by higher consumer-oriented shipments.

Now turning to revenue by business unit. Compute and Networking Business Unit revenue was up 4% sequentially, to $4.6 billion, and reached 57% of our total revenue. For the third consecutive quarter, CNBU revenue reached a new quarterly record, driven by a more than 50% sequential increase in HBM revenue.

Revenue for the Storage Business Unit was $1.4 billion, down 20% sequentially. Decline in SBU revenue was driven primarily by lower storage investments from data center customers after several quarters of very strong growth and overall NAND industry pricing.

Mobile Business Unit revenue was $1.1 billion, down 30% sequentially, as mobile customers continued to improve their inventory positions.

Embedded Business Unit revenue was $1 billion, down 3% sequentially. Lower sequential revenue was primarily due to inventory improvement initiatives at automotive customers.

The consolidated gross margin for fiscal Q2 was 37.9%, down 160 basis points sequentially, due primarily to pricing in consumer-oriented segments of the market, especially in NAND, and NAND mix shift to consumer-oriented products, as mentioned earlier. Our ongoing high-value mix shift in our DRAM portfolio partially offset some of these factors.

Operating expenses in fiscal Q2 were $1 billion, flat sequentially. R&D expenses were lower than planned due to earlier product qualification and timing of certain R&D projects.

We generated operating income of $2 billion in fiscal Q2, resulting in an operating margin of 24.9%, which was down approximately 260 basis points sequentially and up 21 percentage points from a year-ago quarter. Fiscal Q2 adjusted EBITDA was $4.1 billion, resulting in an EBITDA margin of 50.7%, up 10 basis points sequentially and up 14 percentage points or $2 billion from the year-ago quarter.

Fiscal Q2 taxes were $214 million on an effective tax rate of 10.7%, lower than our guidance due to the effects of one-time items in the quarter.

Non-GAAP diluted earnings per share in fiscal Q2 was $1.56, above the high-end of the guidance range, compared to $1.79 per share in the prior quarter and $0.42 in the year-ago quarter.

Turning to cash flows and capital spending. In fiscal Q2, our operating cash flows were over $3.9 billion and our capital expenditures were $3.1 billion, net of proceeds from government incentives. As a result, free cash flows in the quarter were $857 million.

Our fiscal Q2 ending inventory was $9 billion or 158 days, up as communicated previously, and an increase of nine days from the prior quarter.

On the balance sheet, we held $9.6 billion of cash and investments at quarter end and maintained $12.1 billion of liquidity when including our untapped credit facility. During fiscal Q2, we extended our debt maturities through a $1 billion 10-year senior note offering and a $1.7 billion term loan, with proceeds principally used to pay down notes maturing in 2026 and the previous term loan balance. We ended the quarter with $14.4 billion in total debt, low net leverage, and a weighted average maturity on our debt of 2032.

Following quarter end, we renewed and increased the size of our five-year revolving credit facility to $3.5 billion. This provides an additional $1 billion of liquidity and further improves our financial flexibility.

Now turning to our outlook for the third fiscal quarter. We forecast growth in DRAM and NAND bit shipments in fiscal Q3. We forecast sequentially lower fiscal Q3 gross margin, which includes the effects of higher consumer-oriented volumes. NAND underutilization continues to weigh on gross margins.

We project operating expenses in fiscal Q3 to be approximately $1.13 billion, and fiscal 2025 OpEx to increase by over 10%, reflecting planned increases to support our portfolio of high-value products, including HBM.

We expect DIO to decrease in the third fiscal quarter on higher bit shipments. We continue to project ending fiscal 2025 with tight DRAM inventories.

For fiscal Q3 and Q4, we estimate our non-GAAP tax rate to be approximately 14%. In fiscal Q3, we forecast CapEx to be over $3 billion. Our CapEx projection for fiscal 2025 remains approximately $14 billion. The overwhelming majority of the fiscal 2025 CapEx is to support HBM, as well as facility, construction, backend manufacturing, and R&D investments.

Impacts from potential new tariffs are not included in our guidance, given the uncertainty around tariff timing, nature, and implementation.

With all these factors in mind, our non-GAAP guidance for fiscal Q3 is as follows: we expect revenue to be $8.8 billion, plus or minus $200 million; gross margin to be in the range of 36.5%, plus or minus 100 basis points; and operating expenses to be approximately $1.13 billion, plus or minus $15 million.

As mentioned, we expect the fiscal Q3 tax rate to be around 14%. Based on a share count of approximately 1.14 billion shares, we expect EPS to be $1.57 per share, plus or minus $0.10.

In fiscal Q2, Micron delivered earnings above guidance range, achieved record revenues again in data center DRAM, ramped our leading HBM, and released the industry's most advanced DRAM process technology. For fiscal Q3, we project record quarterly revenue at the midpoint of our guidance. We are focusing our R&D resources, exercising capital discipline, and maintaining a strong balance sheet as we extend our leadership and tap into substantial growth opportunities ahead.

I will now turn it back over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Mark.

Micron is uniquely positioned to capitalize on the transformative growth driven by AI, from data center to edge devices, and we are on track for the record revenue and significantly improved profitability in fiscal 2025. We are confident in our ability to navigate the current market dynamics with disciplined investments and a focus on our high-value portfolio mix shift. This is the most exciting time I have seen for memory and storage, and Micron's innovations are at the forefront of this revolution. We are excited about the opportunities ahead and remain committed to delivering value for all our stakeholders.

Thank you for joining us today. We will now open for questions.

Questions And Answers

Operator

(Question And Answer)

Certainly. And our first question for today comes from the line of Harlan Sur from J. P. Morgan. Your question please.

Q - Harlan Sur {BIO 6539622 <GO>}

Yes. Good afternoon. Thanks for taking my question. Back in mid-February at an investor conference, I know the team had walked us through the dynamics on a weaker gross margin profile. Here in the May quarter, that's playing out, but you did anticipate an improved gross margin profile beyond this quarter fiscal Q3. So is that still the case that we should see gross margin improvements maybe starting in fiscal Q4 and potentially beyond? And is that across both data center and your consumer-related products? Is that across total DRAM and your NAND segments? Any color here would be great.

A - Mark Murphy {BIO 16136048 <GO>}

Sure, Harlan. This is Mark. I'll take that. So let me just make some comments about the third quarter. It is down sequentially as we had indicated in the conference. And again, as we said in the conference, down primarily due to higher mix of consumer-oriented volumes, lower CQ1 pricing on consumer-oriented markets and industry NAND conditions generally. All that partially offset by higher HBM. Yes, we do see while down, conditions have improved since those public comments. And the updated view is reflected in the guide today.

Now, we're not providing guidance on the fourth quarter. However, we do expect gross margin to be up somewhat. There's always tailwinds and headwinds, as you know. On tailwinds, we do expect market conditions to improve. We do expect HBM and other high-value products to grow and contribute to mix improvement. Some headwinds, we do see NAND underutilization as we talked about. And actually, since our capacity has come down structurally, we're going to see less of those costs in the third quarter on period and see more of those costs hit us as inventory clears in the fourth quarter. It's still -- we've taken actions to manage the NAND supply, and that's important. And that part of the business is still getting its legs back under it, but we intend to take price action in the second quarter calendar and just maintain -- work to maintain supply discipline.

And then we are going to see in the fourth quarter the beginning of some startup costs related to construction activities and new node and DRAM that we're working. So in short, we would expect fourth quarter margins to be up somewhat from third quarter.

Q - Harlan Sur {BIO 6539622 <GO>}

I appreciate that. And then, Sanjay, you increased your industry bit demand outlook from mid-teen last earnings to mid to high-teens this quarter for calendar '25 for DRAM. I think part of it is the HBM-related dynamics as you increased your TAM outlook for HBM this year. But is the team seeing any other segments within DRAM that are driving the better industry bit demand profile as the year unfolds?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

We had projected that customer inventories will get in a better place by spring timeframe in the consumer side of the business. And it's turning out to be the way we had projected. And of course, the smartphone and PC markets are also seeing more and more devices that have AI implemented that drive content growth. So as customer inventories got closer to healthier levels, we are seeing resumption of purchases by customers and all that plays into our guide for 2025 bit demand.

And as you noted, of course, data center continues to be strong. And in data center, of course, HBM is a strong contributor toward revenue growth. In terms of bit demand growth for data center, of course, high-density DIMMs as well as LP where Micron leads the industry, all of these actually contributing toward the demand increase in 2025 as well.

Q - Harlan Sur {BIO 6539622 <GO>}

Thank you, Sanjay. Thank you, Mark.

Operator

Thank you. And our next question comes from the line of Timothy Arcuri from UBS. Your question please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. Mark, can you give us a little detail on the fiscal Q3 guidance? You're guiding revenue up about $750 million. How much of that's coming from DRAM versus NAND? And I know you said that bits are up in both, but can you give us a sense of how much bits are up in each of those two markets? Thanks. And I had another question as well.

A - Mark Murphy {BIO 16136048 <GO>}

Yes, Tim, we've provided you the consolidated revenue number. You have the year-to-date figures on both revenue and bit growth and price for both DRAM and NAND. And then we've provided you a demand growth for the year in bits. So I think we've provided you the contours of the business and you should be able to make some volume and price assumptions on the revenue outlook. We do expect bit growth in both DRAM and NAND in third quarter.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay. But I guess, Mark, can you say that you expect revenue growth in both DRAM and NAND as well?

A - Mark Murphy {BIO 16136048 <GO>}

DRAM with the HBM and data center exposure will be -- the bias of growth will be there.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay. Cool. Thanks. And then -- so, Mark, I mean, obviously everything you're doing here is great. I think the bugaboo, obviously, is margins are still a bit low, certainly into the fiscal Q4. So I guess I have, like, a two-part question. One, I know you don't want to guide fiscal Q4 margins, but do you think you can get back to what you just did in fiscal Q2, in fiscal Q4?

And I guess broadly, when does this stuff clear and sort of we begin to see the true goodness flowing through from HBM and from all the cost downs you're doing on the non-HBM DRAM side? So sort of when do we start to get kind of a clean gross margin number, if you will? Thanks.

A - Mark Murphy {BIO 16136048 <GO>}

Yes, Tim, we're not going to provide a fourth quarter number. We have indicated that fourth quarter gross margins would be up somewhat from third quarter. I think, as Sanjay mentioned in prepare remarks, we're in the best position we've ever been in on technology and market exposure of products. Manufacturing is operating very well.

On the cost side, maybe to help you with the modeling, our all-in DRAM costs for fiscal year '25, we expect to be flattish. Our all-in NAND costs for FY '25, in line with front-end costs, reductions in the low-double-digits. So we are taking supply actions on the NAND side. It consists of underloading, reducing CapEx, delaying node transitions, as we talked about, and we're beginning to see some signs of improvement on that part of the business.

On the DRAM side of the business, we know the continued HBM growth and broader data center, and then the lead edge on DRAM is tight, and we're again projecting our DIO levels to be below our target by the end of the fiscal year.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

And, Tim, I'll just add that, of course, we continue to be focused on increasing the mix of our revenue towards higher profit pools of the industry, both in DRAM and NAND. So in terms of our product portfolio, as we have said, it is best positioned, so continue to drive the product portfolio and mix, focusing on really strengthened profitability. And demand trends, we have talked about. I mean, we feel good about DRAM demand trend, and of course, in NAND, the supply discipline will be important. We, of course, are extremely focused on that, and AI is benefiting our DRAM demand across data center and edge as well. And of course, our technology position, our product position, and our cost position continues to be healthy.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay. Thank you both.

Operator

Thank you. And our next question comes from the line of Krish Sankar from TD Cowen. Your question please.

Q - Krish Sankar {BIO 16151788 <GO>}

Thank you for the question. I have two of them. Firstly, Sanjay and Mark, you've seen some of the memory prices improve off late. I'm just wondering how much of that is true end demand versus actually tariff-related pull-ins, and how sustainable do you think the industry pricing dynamics today are?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So, of course, as we have indicated, the DRAM demand drivers, as well as NAND on the consumer side of the business, in smartphones -- particularly in smartphones, as well as in PCs, are improving as the customers are getting closer to their normal inventory levels in the consumer markets.

Again, along the lines of what we had projected. PC, probably more second half of this year that we start seeing greater demand trend with respect to AI PCs and increasing penetration of AI PCs. AI PCs require greater DRAM content than what was in the past. We have talked about that last year, average 12 gigabyte in PCs and AI PCs with NPU running at 40 TOPS or higher requires 16 gigabyte or higher.

So these are good demand trends on the PC side, and same thing happening on the smartphone side. You have seen several introductions and more of these are smartphones and more of them to be rolling out with AI smartphones that do have higher DRAM content, greater than 12 gigabyte versus last year at 8 gigabyte.

And of course, data center demand trend in DRAM continues to be strong as well. So all of this, I mean, first of all, demand trends are in a good place. And of course, on the supply side, leading edge supply, as we mentioned in the prepared remarks as well as Mark just mentioned, leading edge DRAM supply is tight. And certainly that is happening because of increasing demand for HBM and HBM trade ratio.

And on NAND, the supply actions by various players and underutilization in the fabs certainly is improving the supply picture as well. So all of this is improving the demand-supply environment in the industry. And of course, we are well-focused on driving an inflection toward higher pricing in CQ2. And we are well-positioned with our products across the end market segments. And we really look forward to continue to maximize the opportunities for our business and continue to increase the mix of our business toward higher profit pools of the industry.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it. Got it. Thanks, Sanjay. And then a quick question on the gross margin side. Clearly, HBM3E 8-high is a relatively mature technology as we move to 12-high in HBM4. Would that, the yield might be lower than 8-high? I'm just kind of curious. Does it have any negative impact on gross margins or is it de minimis at this point? Thank you.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

First of all, in HBM3E 8-high, very pleased with how our team has executed. And we mentioned in the prepared remarks that we actually delivered greater volume of HBM3E versus our plans in our FQ2 and exceeded revenue for the first time, a major milestone of more than $1 billion. And HBM3E continues to do well with 8-high. Our yields, our capacity ramp is going well. Our execution is going well. And all that experience of 8-high in terms of capacity ramp as well as yield ramp will of course help us as we ramp our 12-high.

We have announced before that we are now in volume production with our 12-high. Just like any other new product, and these are highly complex products, HBM is the most complex product ever made in the industry, these kind of complex products, of course in the early stages, there is a yield ramp. We expect 12-high to of course have a premium over 8-high and of course will continue to be accretive to our DRAM margins nicely as well.

And we remain very focused in second half, shifting the vast majority of -- second half of the calendar year, shifting the vast majority of our volume to 12-high. And as we ramp that volume, of course yields will continue to go up as well. And with that by the end of this calendar year, as we shared with you, we expect to reach our HBM share to be in line with our overall industry DRAM share.

Q - Krish Sankar {BIO 16151788 <GO>}

Thanks, Sanjay.

Operator

Thank you. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question please.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you. I wanted to make sure I got the inventory targets right. So you're at 153 days and you'll be below the target model, which I think is 120 days in two quarters. And I guess if that's right, that seems like a lot of reduction. Can you talk about how much volume do you need to do that? How much is the inventory kind of impacted by some of the HBM supply trends, things like that? Could you just characterize a little bit how you reduce that much inventory?

A - Mark Murphy {BIO 16136048 <GO>}

Yes, Joe, it's Mark. So 158 days, I think, was the DIO in the second quarter. As we've talked about, conditions are tighter in DRAM than NAND. You've heard about the supply actions that we've taken on NAND. We have seen good volume growth there, and that is expected to continue. But the industry conditions are such that you've heard us talk about underutilization, reducing CapEx, and delaying the node transition.

On the DRAM side, as Sanjay mentioned, AI-driven growth and related to HBM and other product sets, but particularly HBM with the trade ratio is creating tightness in that market. And the target we have for inventories that we've stated before is 120 days, and we would expect to be below that on DRAM in our fiscal fourth quarter.

Q - Joseph Moore {BIO 17644779 <GO>}

Okay. Great. Thank you. And did you guide inventory for this coming quarter? I assume that starts to come down now?

A - Mark Murphy {BIO 16136048 <GO>}

We just said that on days that will go down through the year.

Q - Joseph Moore {BIO 17644779 <GO>}

Yes. Okay. Thank you very much.

Operator

Thank you. And our next question comes from the line of C. J. Muse from Cantor Fitzgerald. Your question please.

Q - C. J. Muse {BIO 6507553 <GO>}

Yes. Good afternoon. Thank you for taking the question. I guess first question, I want to follow-up on gross margins, Mark. Is there a framework for thinking about the underutilization charges and the period cost into kind of May and August on the NAND side? And then is there also a way to think about the incremental construction costs as we go into August and November?

A - Mark Murphy {BIO 16136048 <GO>}

Yes. So C. J., we had less period costs in our third quarter here than we had originally projected, in part due to the fact that we've structurally brought down our capacity. And so more of the underutilization charge will go into inventories and then that will flush through that way versus the period cost.

Those period costs, I'd say the underabsorption costs do weigh on gross margins in fourth quarter and into '26. Now, even with that, the combination of growth in the business, the improved market conditions, and then mixed improvements, those will, we believe, result in somewhat higher gross margins in the fourth quarter.

As it relates to startup costs, the effect is relatively small on a sequential basis, third to fourth quarter. But as we approach wafer outs in Idaho, that number will increase through '26. And so we'll provide more color on that as our plans and timing is finalized.

Q - C. J. Muse {BIO 6507553 <GO>}

Very helpful. And a quick follow-up. You revised your HBM industry revenue outlook higher. Curious if there's a framework in how you're thinking about kind of first half versus second half of the industry. Thanks so much.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Of course, the revenue in the second half, as you go from 8-high to 12-high, continues to go up because 12-high will be carrying certain premium over 8-high. So we have projected more than $35 billion for calendar year 2025, and a bigger portion of that in second half of calendar '25 versus first half.

And more than $35 billion, of course, is the industry TAM for HBM that we have referred to here. And I'll just point out that, of course, with respect to HBM, there is expansion of HBM customer base taking place. Micron itself, now we are shipping to a third large customer that we have begun shipping our products to. So that also is contributing to the growth in HBM revenue in the second half as the customer base expands.

Operator

Thank you. And our next question comes to the line of Chris Caso from Wolfe Research. Your question please.

Q - Chris Caso {BIO 4815032 <GO>}

Yes. Thank you. The first question is with regard to the lower end of the market. And if you could tell us about what the exposure is particularly on the DRAM side, LP4 and DDR4. And I guess as you start to see growth in HBM, get some normalization in some of the consumer markets as you go to the second half of the year, I guess is that going to be kind of have a de minimis effect on revenue and margins as you go into the second half?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So regarding D4 and LP4, last quarter in our earnings call, we had shared that our revenue from those products, LP4 and D4, for the remainder of the fiscal year, we have said at the time, corresponds to about 10% of our total company revenue. So we continue to see that for the remainder of the fiscal year. And your second question is that -- can you repeat the second question?

Q - Chris Caso {BIO 4815032 <GO>}

Yes. It's just -- does that have a de minimis effect on margins? And I guess that the point is, does that cease being a drag on margins as you go in the second half as other parts of the business grow?

A - Mark Murphy {BIO 16136048 <GO>}

I mean, it is a smaller part of the overall company revenue. And it is -- I mean, we are, of course, leading in D5 and HBM and D5-based and LP5-based product in data center as well as other parts of the consumer markets. So of course so, LP4 and D4 will continue to become smaller over time. And just keep in mind that overall industry conditions in DRAM are improving from the point of view of DRAM, demand drivers, and the supply tightness. And that can also play some role in the overall dynamics of all parts of the DRAM market.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you. As a follow-up on HBM, what you said in the past is you're sold out for the year, but yet your TAM assumptions have moved higher for the past several quarters. As we look into next year, presumably the higher TAM this year translates to higher TAM assumptions as you go to next year. Would you still have the capability to increase your capacity to maintain that market share in HBM, which is equal to your overall share of DRAM as you go into next year?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

So, of course, we have said that as we exit calendar '25, our share in HBM will correspond to our industry DRAM share. So if you look at that run rate, I mean, clearly in calendar year 2026, our share would be higher for the full year basis versus 2025. And we remain very focused on continuing to increase our capacity of HBM.

I mentioned earlier that we are doing -- executing quite well in terms of continuing to increase the capacity, continuing to shift from 8-high to 12-high, very much focused on bringing HBM4 into the market next year, and, of course, addressing all the capacity needs related to that as well. So while we are not projecting 2026 market share at this time, we feel very good about our HBM position, our close relationships with our customer, our execution on our technology and products and our manufacturing overall.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you.

Operator

Thank you. And our next question comes from the line of Chris Danely from Citi. Your question please.

Q - Christopher Danely {BIO 3509857 <GO>}

Hey. Thanks, guys. So if we look at the May guidance and then just going back the last three quarters, your revenue is between $7.8 billion and $8.7 billion per year, and the last time that happened was about three years ago, that was the last upturn, but your gross margins were 10 points higher. So why are gross margins -- because depreciation hasn't changed that much, why are gross margins like 10 points lower and essentially the same revenue base? And does this mean that we can forget about your gross margins ever going to the 50%s again? Or maybe give us a path to getting them back there, if you think that that's conceivable.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Overall, our gross margins in DRAM have been healthy and, again, supported by our strong technology and product positions based on our D5 products, LP5 products, HBM products. NAND is what has weighed down on our margins. And of course, we have continued to focus, in NAND, it is because of the overall industry environment and overall industry demand-supply imbalances.

And so, of course, both in DRAM and NAND, it's always a function of demand-supply environment, but also very much a focus of ours on increasing the mix of the business toward high-value solutions. And that's what we continued to do in NAND as well. And as we see a greater supply discipline, we would certainly fully expect that NAND fundamentals would improve in the industry as well. Of course, it's important to maintain the focus on sustained supply discipline there as well.

So as we look ahead, as we have pointed out, that we remain very focused on continuing to strengthen the mix of our revenue, mix of our products in the business toward high-margin products, both in NAND and DRAM, continue to focus on product portfolio strength, managing demand-supply very closely, and managing our technology development and ramp into production closely to make sure that our supply and demand is well-aligned and, of course, very much focused on overall costs as well. And with that, we are certainly optimistic here that the structural overall with the industry that the improvements will occur in the business fundamentals.

Q - Christopher Danely {BIO 3509857 <GO>}

Thanks, Sanjay. Just one brief follow-up on that. So you talked about increasing the mix to higher value solutions. And then in the commentary and also in the press release, you said that part of the reason that the gross margin was lower was this increasing consumer exposure. And your NAND actually was down a lot more than DRAM. So are you seeing increased consumer exposure in DRAM, and why is that happening, and how do we -- or how do you guys change that?

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Well, first of all, we are doing well in data center, and our mix of business and data center for DRAM, as we have pointed out, continues to increase. And we have leadership products in DRAM with data center, including not just HBM, but we have also talked about when we just announced SOCAMM products. These are important products. So doing well with respect to DRAM.

On the consumer side, of course, over the last few quarters, there was overhang of customer inventory on the consumer side, and that was there in DRAM as well. And as that customer inventories get closer to normalization, along the lines of what we have said before, of course, along with the AI drivers in consumer devices, we see a strong bounce back, particularly in smartphones, with respect to demand. And that's causing, actually, overall, HBM trade issue, as well as a strong bounce back in the consumer demand, is causing tightness in leading edge as well. And of course, these fundamentals of demand and supply enable us to drive inflection in pricing higher in CQ2 timeframe.

And same thing on the NAND side that supply actions that have been taken in the industry, as well as consumer inventories -- on the customer/consumer inventories getting normalized is bringing back demand on the NAND side as well. And of course, we'll be driving inflection in pricing higher on the NAND side as well in CQ2.

Q - Christopher Danely {BIO 3509857 <GO>}

Got it. Thanks.

Operator

Thank you. And this does conclude the question-and-answer session as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.
